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Dataset: transaction.csv (Sample of Accounting Transactions)



EXECUTIVE SUMMARY

This automated audit analyzes financial transaction data to detect potential anomalies such as duplicate invoices, negative amounts, missing fields, statistical outliers, weekend transactions, and unusually high-value entries. The goal is to help accounting teams quickly identify data integrity risks and strengthen internal controls.

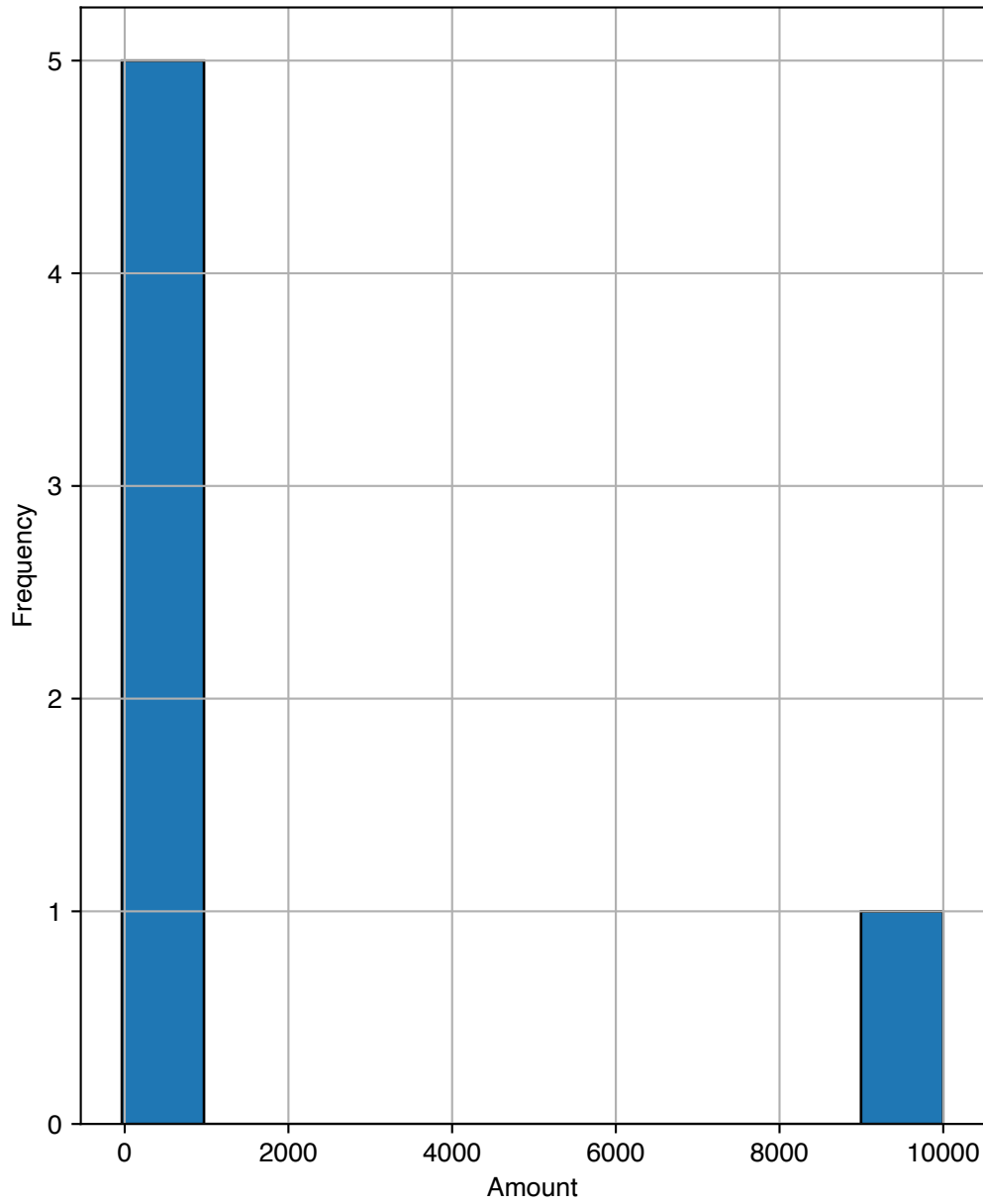
Metric	Count
Total Records	6
Duplicate Invoices	2
Negative Amounts	1
Missing Values	1
Outliers (z-score ≥ 3)	0
Outliers (IQR)	1
Over Threshold ($ Amount > 1000$)	1
Weekend Transactions	2
Empty Vendor	1

Executive Insights

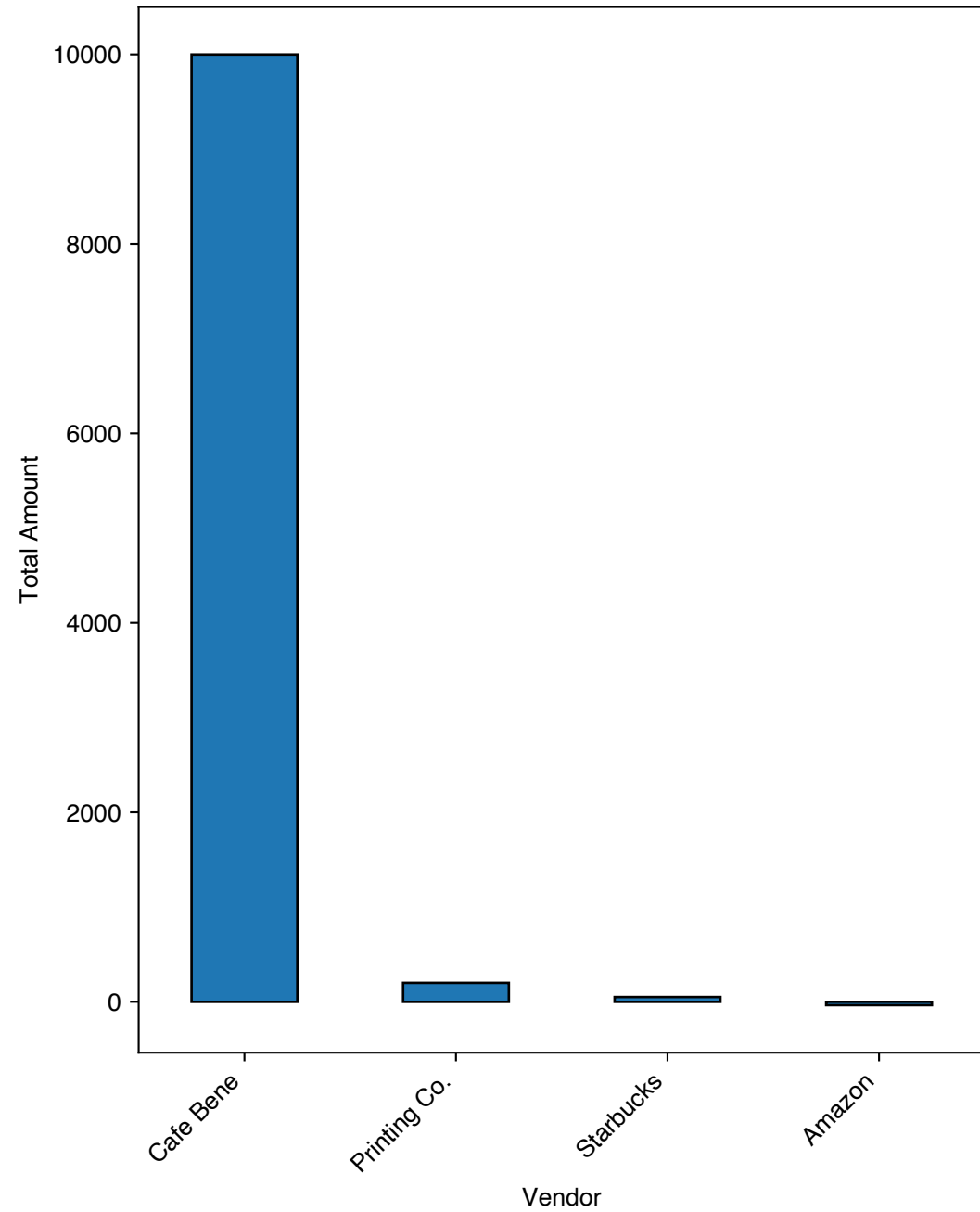
- Weekend-related anomalies: 22.2% of total detected.
- High-value ($|Amount| > 1000$): 11.1% of total detected.
- Top duplicate vendors: Starbucks (2)

Visual Insights

Transaction Amount Distribution



Top Vendors by Total Amount



Top 3 Anomalies

Transactions requiring immediate review.

Invoice_ID: INV001
Vendor: Starbucks | Amount: 25.5
Category: Marketing | Date: 2025-09-01

Invoice_ID: INV001
Vendor: Starbucks | Amount: 25.5
Category: Marketing | Date: 2025-09-08

Invoice_ID: INV002
Vendor: Amazon | Amount: -35.0
Category: Supplies | Date: 2025-09-01

Recommendations

- Enforce mandatory Vendor field validation with dropdowns.
- Validate uniqueness of Invoice_ID before posting.
- Require secondary approval for high-value or weekend transactions.
- Block uploads with critical missing fields; provide error report.
- Monitor statistical outliers monthly and investigate root causes.

Appendix: Anomaly Samples (First 10 Rows Each)

Duplicate Invoices

{'Invoice_ID': 'INV001', 'Date': '2025-09-01', 'Vendor': 'Starbucks', 'Amount': '25.5', 'Category': 'Marketing'}

{'Invoice_ID': 'INV001', 'Date': '2025-09-08', 'Vendor': 'Starbucks', 'Amount': '25.5', 'Category': 'Marketing'}

Negative Amounts

{'Invoice_ID': 'INV002', 'Date': '2025-09-01', 'Vendor': 'Amazon', 'Amount': '-35', 'Category': 'Supplies'}

Missing Values

{'Invoice_ID': 'INV005', 'Date': '2025-09-06', 'Vendor': 'nan', 'Amount': '150', 'Category': 'Equipment'}

Outliers (IQR)

{'Invoice_ID': 'INV006', 'Date': '2025-09-07', 'Vendor': 'Cafe Bene', 'Amount': '9999', 'Category': 'Food'}

Weekend Transactions

{'Invoice_ID': 'INV005', 'Date': '2025-09-06', 'Vendor': 'nan', 'Amount': '150', 'Category': 'Equipment'}

{'Invoice_ID': 'INV006', 'Date': '2025-09-07', 'Vendor': 'Cafe Bene', 'Amount': '9999', 'Category': 'Food'}